

Habib University
CSE 351 – Artificial Intelligence
Fall 2026
Assignment 02
100 Points

Question 1 – Neural Networks as Universal Approximators [35 Points]

One of the most powerful abilities of neural networks is their capacity to serve as universal function approximators. Many real-world problems, such as predicting stock prices, modeling physical systems, or approximating solutions to complex equations, can be expressed as learning an unknown function from input–output examples (training data). A Multi-Layer Feedforward Neural Network (MFNN) can approximate such functions to arbitrary accuracy, given sufficient neurons and training data. This assignment explores the role of MFNNs in function approximation, helping students understand how networks learn mappings, generalize patterns, and handle nonlinear relationships.

Using an MFNN, you are required to approximate the following function:

$$f(x, y) = e^{-\frac{x^2+y^2}{8}} \cos\left(\sqrt{x^2 + y^2} + c\right)$$

where c is a small constant (you can take $c = 1 \times 10^{-3}$). Inputs x and y range over $[-2, +2]$. For the MFNN, use the standard backpropagation algorithm.

Your Python code will do the following:

a) Data Generation

For training data, generate an array of data ranging from -2 to $+2$ with a spacing of 0.2 (you may use `x = np.arange(-2, 2.1, 0.2)` in Python using numpy). Similarly generate the y -vector. Concatenate the two vectors x and y to get a 2D input training data matrix. Corresponding ground-truth (true values) can be found using the formula given above. For testing data, use the same procedure but this time generate data from -1.9 to $+1.9$ with a spacing of 0.2 .

b) Model Training

Do your own implementation of the backpropagation algorithm to train an MFNN. Provide a plot showing the cost function versus epoch. You may use the following hyperparameters:

- Hidden layer = 1
- Number of neurons in the hidden layer = 40
- Choice of activation function: tansig
- Learning rate = 0.05
- Epochs = 5000
- Initial bias value = 0.5

c) Model Testing

Pass test data through the trained network to predict the output. Plot the predicted output alongside the true (ground truth) values. Ensure that training and test data are kept separate, and display the curves for both sets on the plot.

d) Fine-Tuning

Explore the effects of the following hyperparameters on your output and report your findings:

- Number of layers
- Activation function
- Biases
- Learning rates (say 0.01, 0.1, 0.3)

The code will be written in the attached Q1.ipynb.

Question 2 – Decision Tree Induction

[35 Points]

Decision trees are among the most widely used methods in Artificial Intelligence for classification and prediction. They approximate decision-making processes by breaking down complex problems into a sequence of simple choices, represented as branches in a tree. The decision tree can be induced by selecting attributes that maximize the GINI index, enabling the model to discover meaningful patterns in data. This assignment will give you hands-on experience in constructing decision trees and understanding how they approximate decision functions.

You are provided with three datasets, i.e. Data1, Data2, and Data3 (contained in the DATA.mat file). The three datasets are synthetic benchmarks designed to test decision tree learning under different conditions. Data1 is rule-based with simple equality and OR conditions. Data2 is parity-based, requiring recognition of global counting patterns, which makes it the most difficult. Data3 includes both complex rules and 5% noise, providing a test of robustness against imperfect data.

Each dataset consists of a train and test sequence; the last column indicates the class of the sample. The data consists of six discrete-valued attributes taking the following values: $a_1 \in \{1, 2, 3\}$; $a_2 \in \{1, 2, 3\}$; $a_3 \in \{1, 2\}$; $a_4 \in \{1, 2, 3\}$; $a_5 \in \{1, 2, 3, 4\}$; $a_6 \in \{1, 2\}$.

You will develop your decision trees using the train data, and testing will be performed using the test data.

a) GINI Index

Compute the GINI index of the six attributes in the three training datasets. Fill the table given below. Based on the results, which attribute should be used for splitting the examples at the root node for each dataset? Do you find GINI a robust measure for difficult datasets as well?

Dataset	a_1	a_2	a_3	a_4	a_5	a_6
Data1						
Data2						
Data3						

b) Tree Construction

Now consider Data1 only. Develop a complete tree in Python using GINI index recursive partitioning, and find the accuracy of your algorithm using the test samples. Visualize the tree using a graph library (such as networkX or GraphViz).

The code will be written in the attached Q2.ipynb.

Question 3 – Image Classification for Object Recognition [30 Points]

You are given the CIFAR-10 dataset, comprising 32x32 colour images of 60,000 objects from 10 categories, with 6,000 images per category. The training set has 50,000 images and the test set has 10,000 images.

The dataset is available at <https://www.cs.toronto.edu/~kriz/cifar.html> (or via your ML framework of choice, e.g. `tf.keras.datasets.cifar10`).

You have to build a scikit-learn based classifier for this dataset to classify each input image into one (out of 10) categories. You are required to experiment with four different classifiers and report the accuracy of each of them:

- (a) Decision Tree
- (b) Artificial Neural Network
- (c) Support Vector Machine (SVM)
- (d) K-Nearest Neighbour (kNN)

Note: raw-pixel SVM/kNN training can be slow on the full 50,000-image training set. You may train on a representative subsample (e.g. 5,000–10,000 images) if needed for runtime – state your choice and justify it briefly in your report.

The code will be written in the attached `Q3.ipynb`.

Submission Instructions

The assignment will be submitted on LMS by the due date (announced on LMS). No email submission will be accepted. The submitted file should be in the form of a ZIP file named as `<studentid1>_<studentid2>_HW2` containing separate folders for each question. The skeleton code is provided for all three questions. Additionally, there should be a single PDF file containing results, analysis, and plots, etc. for all questions.